Report

Shortest Route from Water Depot to a Repair Location using Dijkstra's Algorithm

1. Problem Statement

Develop a C++ program to represent a water depot, road junctions, and repair locations as vertices with non-negative travel distances as edge weights. Apply Dijkstra's algorithm to determine the least-cost route to a selected repair location and display the route and total distance. Modify the program to (A) select a different source vertex and recalculate the shortest distances, (B) display the complete shortest path to a destination, not only its distance, and (C) update an edge weight and recalculate the shortest paths.

2. Objective

To represent a water supply road network as a weighted graph, with the water depot, road junctions and repair locations as vertices and roads as edges, and apply Dijkstra's algorithm to find the least-cost route and complete path from a selected source to a repair location, recalculate the shortest distances for a different source vertex, update a road distance and recalculate the shortest paths, handling invalid locations and unreachable routes.

3. Algorithm

Step 1: START.

Step 2: Build the graph as an adjacency list: each water depot, road junction and repair location is a vertex and each road is an undirected edge (u, v) with a non-negative weight w (distance in km).

Step 3: Read the source location and the repair location. If either location is not present in the graph, or the destination is not a repair location, display an error and STOP.

Step 4: For every location, initialize dist[] = INFINITY, parent[] = -1 and visited[] = false.

Step 5: Set dist[source] = 0 and push (0, source) into a min-priority queue.

Step 6: WHILE the priority queue is not empty, repeat Steps 7 to 9. When it becomes empty, go to Step 10.

Step 7: Remove the pair (d, u) with the smallest distance from the queue. If visited[u] = true, skip it and go back to Step 6.

Step 8: Mark visited[u] = true.

Step 9: For each unvisited neighbour v of u with edge weight w: if dist[u] + w < dist[v], then set dist[v] = dist[u] + w and parent[v] = u, and push (dist[v], v) into the queue.

Step 10: If dist[repair location] = INFINITY, display "No route exists" and go to Step 13.

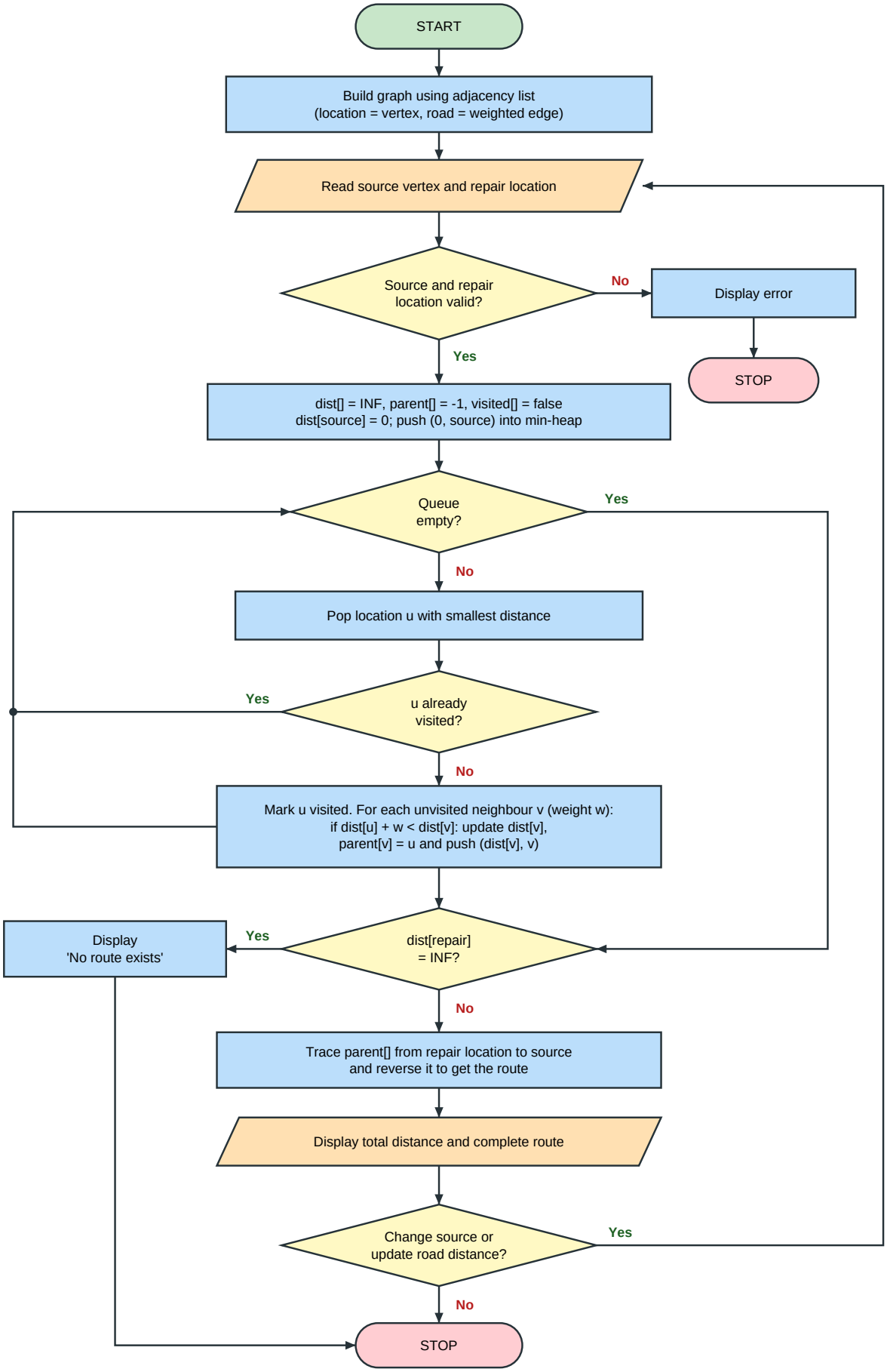
Step 11: Build the route: starting from the repair location, follow parent[] until the source is reached (parent = -1), then reverse the sequence.

Step 12: Display the total distance and the complete route to the repair location, and the distance and complete route to every other location (Modification B).

Step 13: Ask the user to choose: (a) change the source location (Modification A): read the new source, keep the repair location and repeat Steps 4 to 12; (b) update a road distance (Modification C): read the two locations and the new distance, reject a negative distance or a missing road, otherwise update the weight in both directions and repeat Steps 4 to 12 from the same source; (c) exit.

Step 14: STOP.

4. Flowchart


START
Build graph using adjacency list
(location = vertex, road = weighted edge)
Read source vertex and repair location
Source and repair
location valid?
No
Display error
STOP
Yes
dist[] = INF, parent[] = -1, visited[] = false
dist[source] = 0; push (0, source) into min-heap
Queue
empty?
Yes
No
Pop location u with smallest distance
u already
visited?
Yes
No
Mark u visited. For each unvisited neighbour v (weight w):
if dist[u] + w < dist[v]: update dist[v],
parent[v] = u and push (dist[v], v)
Display
'No route exists'
Yes
dist[repair]
= INF?
No
Trace parent[] from repair location to source
and reverse it to get the route
Display total distance and complete route
Change source or
update road distance?
Yes
No
STOP